

CENTRAL FORCE SYSTEM

$$\Rightarrow \Psi(\vec{r}) = \Psi(r, \theta, \phi) = R(r) \Theta(\theta) \Phi(\phi)$$

Lap. spherical coordinates

$$* \nabla^2 = \frac{d^2}{dr^2} + \left(\frac{2}{r}\right) \frac{d}{dr} + \frac{1}{r^2} \left[\frac{d^2}{d\theta^2} + \cot\theta \frac{d}{d\theta} + \csc^2\theta \frac{d^2}{d\phi^2} \right]$$

Schrödinger Eq.

$$\underbrace{-\frac{\hbar^2}{2m} \nabla^2 R\Theta\Phi + U R\Theta\Phi = E R\Theta\Phi}_{(*)}$$

$$(*) \Rightarrow -\frac{\hbar^2}{2m} \left\{ \frac{d^2}{dr^2} R\Theta\Phi + \left(\frac{2}{r}\right) \frac{d}{dr} R\Theta\Phi + \frac{1}{r^2} \frac{d^2}{d\theta^2} R\Theta\Phi + \frac{1}{r^2} \cot\theta \frac{d}{d\theta} R\Theta\Phi + \frac{1}{r^2} \csc^2\theta \frac{d^2}{d\phi^2} R\Theta\Phi \right\}$$

dividing by $R\Theta\Phi$

$$\begin{aligned}
 (*) : \frac{-\hbar^2}{2m} \left\{ \frac{\Theta \Phi}{R \Theta \Phi} \frac{\partial^2 R}{\partial r^2} + \frac{2}{r} \frac{\Theta \Phi}{R \Theta \Phi} \frac{\partial R}{\partial r} \right. \\
 + \frac{1}{r^2} \frac{R \Phi}{R \Theta \Phi} \frac{\partial^2 \Theta}{\partial \theta^2} + \frac{1}{r^2} \cot \theta \frac{R \Phi}{R \Theta \Phi} \frac{\partial \Theta}{\partial \theta} \\
 \left. + \frac{1}{r^2} \csc^2 \theta \frac{R \Theta}{R \Theta \Phi} \frac{\partial^2 \Phi}{\partial \phi^2} \right\}
 \end{aligned}$$

Then..

$$\frac{-\hbar^2}{2m} \frac{1}{R} \left\{ \frac{\partial^2 R}{\partial r^2} + \frac{2}{r} \frac{\partial R}{\partial r} \right\}$$

$$\frac{-\hbar^2}{2m r^2 \Theta} \left\{ \frac{\partial^2 \Theta}{\partial \theta^2} + \cot \theta \frac{\partial \Theta}{\partial \theta} \right\}$$

$$\left\{ \begin{aligned} & \csc x \\ & = \frac{1}{\sin x} \end{aligned} \right\}$$

$$\frac{-\hbar^2}{2m} \frac{\csc^2 \theta}{r^2} \frac{1}{\Phi} \frac{\partial^2 \Phi}{\partial \phi^2} + \frac{U R \Theta \Phi}{R \Theta \Phi} = E \frac{R \Theta \Phi}{R \Theta \Phi}$$

$U(r, \theta, \phi) \rightarrow U(r)$ (Puerza central).

Isolating dependance on ϕ :

→ Multiply by $r^2 \sin^2 \theta$

$$-\frac{\hbar^2}{2m} \frac{r^2 \sin^2 \theta}{R} \left\{ \frac{d^2 R}{dr^2} + \frac{2}{r} \frac{d}{dr} R \right\}$$

$$-\frac{\hbar^2}{2m} \frac{\sin^2 \theta}{\Theta} \left\{ \frac{d^2 \Theta}{d\theta^2} + \cot \theta \frac{d\Theta}{d\theta} \right\}$$

$$-\frac{\hbar^2}{2m} \frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = [E - U(r)] r^2 \sin^2 \theta$$

Entonces...

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = - \frac{r^2 \sin^2 \theta}{R} \left\{ \frac{d^2 R}{dr^2} + \frac{2}{r} \frac{d}{dr} R \right\}$$

$$- \frac{\sin^2 \theta}{\Theta} \left\{ \frac{d^2 \Theta}{d\theta^2} + \cot \theta \frac{d\Theta}{d\theta} \right\}$$

$$- \frac{2m}{\hbar^2} [E - U(r)] r^2 \sin^2 \theta$$

Así,

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = - \sin^2 \theta \left\{ \frac{r^2}{R} \left[\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{d}{dr} R \right] \right.$$

$$+ \frac{1}{\Theta} \left[\frac{d^2 \Theta}{d\theta^2} + \cot \theta \frac{d\Theta}{d\theta} \right]$$

$$+ \frac{2m}{\hbar^2} [E - U(r)] \left. \right\}$$

For the left-side Eq.

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi} = -m_\ell^2$$

$$\sin \rightarrow \Phi(\phi) = e^{im_\ell \phi}$$

m_ℓ entero

For the right-side Eq.

$$-\sin^2 \theta \frac{r^2}{R} \left[\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{d}{dr} R \right]$$

$$-\frac{\sin^2 \theta}{\theta} \left[\frac{d^2 \theta}{d\theta^2} + \cot \theta \frac{d\theta}{d\theta} \right]$$

$$-\sin^2 \theta \frac{2mr^2}{\hbar^2} [E - U(r)] = -m_e^2$$

Separating variables

$$\frac{r^2}{R} \left[\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{d}{dr} R \right] + \frac{2mr^2}{\hbar^2} [E - U(r)]$$

$$= m_e^2 / \sin^2 \theta - \frac{1}{\theta} \left[\frac{d^2 \theta}{d\theta^2} + \cot \theta \frac{d\theta}{d\theta} \right]$$

Right-side Eq. (2):

$$\left[\frac{d^2 \Theta}{d\theta^2} + \cot \theta \frac{d\Theta}{d\theta} \right] - \Theta m_l^2 \csc^2 \theta = -\Theta l(l+1)$$

↳ $\sin \theta \Rightarrow \Theta(\theta) \rightarrow$ Associated Legendre Polynomial

Finally, for the left-side

$$\frac{1}{R} \left[\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right] + \frac{2mr^2}{\hbar^2} [E - U(r)] = l(l+1)$$

$$\frac{\hbar^2}{2m} \left[\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right] + E(r) R - U(r) R = \frac{\hbar^2}{2mr^2} l(l+1) R$$

$$\frac{\hbar^2}{2m} \left[\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right] + U(r) R(r) + \frac{l(l+1)\hbar^2}{2mr^2} R(r) = E(r) R(r)$$

Radial Wave Equation.